

Automatic Detection of Lexical Ambiguity Patterns in Functional Requirements: An Exploratory Study in the Meat Traceability Domain

Ariel Omar Castro Bajaan, Kleber Obed Crespo Espinoza, Edhu Xavier Gamarra Araujo,
Carlos Andrés Pérez Ruiz, and Erick Jahir Quintero Gende

*Facultad de Ciencias de la Computación, Universidad Técnica Estatal de Quevedo (UTEQ), Quevedo,
Los Ríos, Ecuador*

{acastrob, krespoe, egamarraa, cperezr3, equinterog}@uteq.edu.ec

Abstract. Context. Functional requirements (FRs) written in natural language are prone to lexical ambiguity, propagating defects to later development phases. Automated detectors for Spanish-language FR corpora in Latin American agroindustrial domains are scarce. **Objective.** To characterise the lexical-syntactic ambiguity patterns present in the 27 FRs of CarniCore—a real-world meat distribution management system in Ecuador—using a rule-based automatic detector. **Method.** A deterministic Python detector covering three categories (vague quantifiers, multiple conjunctions, and agentless passive voice) was applied to the complete verbatim FR corpus of ERS/SRS v2.0. The study protocol was pre-registered at OSF prior to data collection [4]. **Results.** One out of 27 FRs (3.7%) triggered the detector: RF-27 activated the vague-quantifier category through the term *correspondiente*. The remaining 26 FRs (96.3%) showed no lexical-syntactic patterns. **Conclusions.** Rule-based detectors act as a low-cost first filter for FR quality; a near-zero rate shifts residual ambiguity risk to the semantic and pragmatic levels. Replication package: <https://doi.org/10.5281/zenodo.22225854>.

Keywords: Requirements ambiguity · Automatic detection · Functional requirements · Empirical study · Ecuador · Meat traceability

1 Introduction

Meat distribution companies in Ecuador manage animal traceability manually, generating inconsistencies that hinder compliance with sanitary regulations imposed by Agrocalidad. CarniCore digitalises that process through 27 functional requirements written in natural-language Spanish, covering supplier registration, animal intake, cold-chain monitoring, weight ticketing, and QR-based traceability [11]. The lexical quality of those FRs determines downstream quality: ambiguous requirements propagate defects to design, testing, and maintenance [7].

Automatic ambiguity detection has been studied mainly for English-language corpora [6, 8, 10]. Cheng et al. [5] identify this geographic and linguistic gap in their systematic review of 100+ studies: no work targets Spanish FR corpora from Latin American agroindustrial domains. Arora et al. [1] confirm that most detectors were validated on software products from high-income countries, limiting generalisability.

This work makes three contributions: (1) an open-source rule-based detector of lexical-syntactic ambiguity patterns in Spanish; (2) its application to a real corpus of 27 FRs of a meat traceability system, yielding the first documented characterisation of this domain in the requirements ambiguity literature; and (3) a FAIR replication package with anonymised data and reproducible scripts.

RQ1. What proportion of CarniCore’s FRs triggers at least one lexical-syntactic ambiguity pattern?

RQ2. Which ambiguity categories are most frequent in the corpus?

2 Related Work

Table 1 summarises the six most relevant studies. Ferrari and Esuli [8] achieved $F_1 \geq 0.70$ using NLP cross-domain models for English. Femmer et al. [6] introduced 12 requirements smells detectable by rules, applied to industrial German-language documents. Gervasi et al. [10] provided a taxonomic review but no applied detector. Fischbach et al. [9] combined NLP and LLMs to extract acceptance tests. Unlike all prior work, this study targets Spanish, an unexplored agroindustrial domain, and makes the corpus and detector fully reproducible under CC BY 4.0.

Table 1: Comparison with related work on FR ambiguity detection

Reference	Year	Lang.	Method	Key difference
Ferrari & Esuli [8]	2019	EN	NLP cross-domain	No Spanish, no agro
Femmer et al. [6]	2017	DE/EN	Rules (smells)	No Spanish corpus
Gervasi et al. [10]	2017	EN	Taxonomy review	No applied detector
Fischbach et al. [9]	2023	EN	NLP + LLM	Acceptance tests only
Cheng et al. [5]	2026	EN	SLR (100+ studies)	No applied detector
This work	2026	ES	Rules	Spanish, meat, FAIR

3 Methodology

3.1 Study Design and PICOC

We conducted a descriptive exploratory study [14]: **Population**—27 verbatim FRs from ERS/SRS v2.0 of CarniCore (Pucayacu, La Maná, Ecuador); **Intervention**—rule-based automatic detector; **Comparison**—none (single-arm; expert panel deferred to future work, see DEV-01); **Outcome**—proportion of FRs triggering ≥ 1 category; **Context**—4th-year undergraduate project, UTEQ, 2026.

3.2 Detector Design

The detector (`detector_ambiguedad.py`) applies three lexical-syntactic rules over the verbatim Spanish text of each FR:

C1 – Vague quantifiers. Matches terms whose referent is unverifiable without additional context: *algunos*, *varios*, *adecuado*, *suficiente*, *rápido*, *correspondiente(s)*, *oportuno*, *apropiado*, *razonable*, among others.

C2 – Multiple conjunctions. Flags FRs containing > 3 connectors (*y/o*) or > 2 modal verbs, following Femmer et al.’s compound-sentence smell [6].

C3 – Agentless passive voice. Detects participial constructions (*será generado*, *deberá ser registrado*) without an explicit actor, violating verifiability [11].

A FR receives `ambiguo_detector = 1` if it activates ≥ 1 category. The analysis is fully deterministic and reproducible from the raw corpus file [3].

3.3 Participants and Data Collection

Field work was conducted in two rounds (July–August 2026) under the LOPDP [2]. Five semi-structured interviews were conducted with four stakeholder profiles: warehouse manager ($n=2$), butcher/dispatcher ($n=1$), reception and weighing operator ($n=1$), and owner ($n=1$). A questionnaire was applied to 31 respondents across six operational roles (weighing and labelling: 7; supplier reception: 7; sales: 6; butchering: 5; supervision: 3; storage: 2). Five walkthrough sessions validated the FR specification. Identifiable material is stored in an AES-256 encrypted container accessible only to the supervising faculty member; only anonymised data appear in the public package [3].

3.4 Protocol Registration and Deviations

The protocol was pre-registered at OSF on 02/08/2026, before data collection, following Nosek et al. [12]. Two deviations are documented: **DEV-01**—the comparative phase (precision, recall, F_1 , Cohen’s κ against an expert panel) was not executed due to the unavailability of independent evaluators; no metrics are imputed. **DEV-02**—the corpus was extended from 25 to 27 FRs after CCB approval of RF-26 and RF-27 post-registration.

4 Results

4.1 RQ1: Proportion of Flagged FRs

Table 2 shows the output of `run_all.sh` over the 27-FR corpus. One FR out of 27 (3.7%) was flagged by the detector.

Table 2: Detector results across the 27 FRs of CarniCore

Category	FRs triggered	Percentage
C1 – Vague quantifier	1 / 27	3.7%
C2 – Multiple conjunctions	0 / 27	0.0%
C3 – Agentless passive voice	0 / 27	0.0%
Any category (flagged)	1 / 27	3.7%

The sole flagged requirement is RF-27:

*“El sistema deberá permitir . . . mostrando los totales **correspondientes** a ese proveedor en el rango de fechas seleccionado.”*

The term *correspondientes* is a vague quantifier: it does not specify which totals are included (sales only? with returns? before or after tax?), making RF-27 unverifiable without stakeholder clarification [13].

4.2 RQ2: Category Distribution

Figure 1 shows the distribution of activated categories. Figure 2 maps detection status per individual FR. Categories C2 and C3 recorded zero activations, indicating consistent use of simple conjunctions and active voice (*El sistema deberá. . .*) across the corpus.

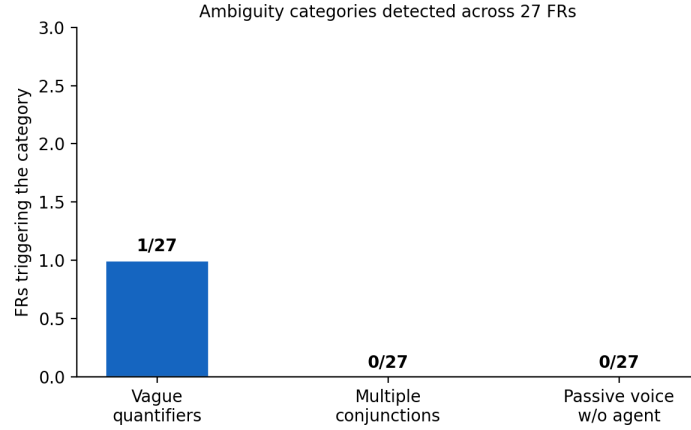


Figure 1: Distribution of detected ambiguity categories ($n=27$ FRs). Generated by `run_all.sh` from `clasificaciones_detector.csv`.



Figure 2: FR-level detection status: green = not flagged (26/27); red = flagged (1/27, RF-27).

5 Discussion

5.1 Implications for Research

The near-zero detection rate suggests that the active-voice template combined with explicit quantitative thresholds (“máximo 3 días para pollo”) acts as an effective embedded lexical-quality control, consistent with Ferrari and Esuli [8], who report lower ambiguity rates in corpora produced with structured elicitation protocols. For Spanish-language corpora produced with similar templates, semantic and pragmatic ambiguity are likely more prevalent quality risks than lexical ambiguity.

5.2 Implications for Practice

Two actionable insights emerge. First, enforcing an active-voice template with mandatory thresholds at writing time reduces lexical ambiguity without post-hoc tooling. Second, the sole flagged term (*correspondientes* in RF-27) is remediable: replacing it with “los totales de ventas, compras y pérdidas netas antes de devoluciones” removes the ambiguity without changing intent.

5.3 Surprising Findings

The complete absence of C2 and C3 across 27 FRs is counterintuitive given the complexity of some requirements (e.g., RF-08 specifies four shelf-life limits in one sentence; RF-21 describes an offline-sync mechanism). We attribute this to the modal anchor *deberá permitir*, which naturally forces active voice and limits connector proliferation.

6 Threats to Validity

Following Wohlin et al. [15]:

Internal. The analysis is deterministic and reproducible; no selection bias (complete census). Threat: regex coverage—terms absent from the pattern list would not be detected.

Construct. The detector covers only lexical-syntactic ambiguity; semantic and pragmatic ambiguity require expert reading.

External. 27 FRs from one system in one country; generalisation requires replication. The published dataset enables this [3].

Conclusion. Without an expert panel (DEV-01), precision and recall cannot be computed. Results are descriptive only.

7 Conclusions and Future Work

RQ1: 1 out of 27 FRs (3.7%) triggered the detector, well below rates reported for unstructured English corpora. **RQ2:** Only the vague-quantifier category was activated (RF-27, *correspondientes*); multiple conjunctions and agentless passive voice recorded zero activations. Structured active-voice templates with explicit thresholds appear to be effective lexical quality controls for Spanish-language FRs in agroindustrial domains.

Future work: (1) Execute with ≥ 3 expert evaluators (completing DEV-01). (2) Extend to other Spanish-language FR corpora. (3) Compare with LLM-based detection on the same published corpus.

Data Availability

Replication package (anonymised transcripts, questionnaire responses [$n=31$], labelled RF corpus in JSON, traceability matrix, LLM prompts, Python scripts): <https://doi.org/10.5281/zenodo.22225854> (CC BY 4.0). Pre-registration: <https://osf.io/wud69>. Repository: https://github.com/EdhuXav/CarniCore_Proyecto_ISR401.

Use of AI-assisted Technologies

Claude (Anthropic, claude-sonnet-4-6) was used for paragraph editing and $\LaTeX$ formatting. All empirical results and conclusions are the authors' own work validated against primary evidence.

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References

- [1] Chetan Arora, John Grundy, and Mohamed Abdelrazek. Advancing requirements engineering through generative AI: Assessing the role of LLMs. *Communications of the ACM*, 67(6):84–91, 2024. <https://doi.org/10.1145/3626253>.
- [2] Ley orgánica de protección de datos personales, 2021. Registro Oficial Suplemento No. 459.
- [3] Replication package for automatic detection of lexical ambiguity patterns in functional requirements: an exploratory study in the meat traceability domain, 2026. Zenodo.
- [4] <https://osf.io/wud69>, Pre-registration: Automatic detection of ambiguity in functional requirements of CarniCore. <https://osf.io/wud69>, 2026.
- [5] Haoyu Cheng, Jati H. Husen, Yang Lu, Teeradaj Racharak, Nobukazu Yoshioka, Naoyasu Ubayashi, and Hironori Washizaki. Generative AI for requirements engineering: A systematic literature review. *Software: Practice and Experience*, 56(2):141–170, 2026. <https://doi.org/10.1002/spe.3350>.
- [6] Henning Femmer, Daniel Méndez Fernández, Stefan Wagner, and Sebastiano Eder. Rapid quality assurance with requirements smells. *Journal of Systems and Software*, 123:190–213, 2017. <https://doi.org/10.1016/j.jss.2016.10.012>.
- [7] Daniel Méndez Fernández, Stefan Wagner, et al. Naming the pain in requirements engineering. *Empirical Software Engineering*, 22(5):2298–2338, 2017. <https://doi.org/10.1007/s10664-016-9451-7>.
- [8] Alessio Ferrari and Andrea Esuli. An NLP approach for cross-domain ambiguity detection in requirements engineering. *Automated Software Engineering*, 26(3):559–598, 2019. <https://doi.org/10.1007/s10515-019-00261-7>.
- [9] Jannik Fischbach, Julian Frattini, Andreas Vogelsang, Daniel Mendez, et al. Automatic creation of acceptance tests by extracting conditionals from requirements: NLP approach and case study. *Journal of Systems and Software*, 197:111549, 2023. <https://doi.org/10.1016/j.jss.2022.111549>.
- [10] Proc. 25th IEEE RE Conference. Proceedings of the 25th IEEE International Requirements Engineering Conference, year = 2017, pages = 128–137, doi = 10.1109/RE.2017.67, Ambiguity in requirements engineering: A systematic literature review. Proc. 25th IEEE RE Conference. Proceedings of the 25th IEEE International Requirements Engineering Conference, year = 2017, pages = 128–137, doi = 10.1109/RE.2017.67.
- [11] ISO/IEC/IEEE. ISO/IEC/IEEE 29148:2018 — systems and software engineering — life cycle processes — requirements engineering. Tech. rep., ISO, 2018.
- [12] Brian A. Nosek et al. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. <https://doi.org/10.1073/pnas.1708274114>.
- [13] Klaus Pohl. *Requirements Engineering: Fundamentals, Principles, and Techniques*. Springer, 2010.

- [14] Per Runeson and Martin Höst. Guidelines for conducting and reporting case study research in software engineering. *Empirical Software Engineering*, 14(2):131–164, 2009. <https://doi.org/10.1007/s10664-008-9102-8>.
- [15] Claes Wohlin, Per Runeson, Martin Höst, et al. *Experimentation in Software Engineering*. Springer, 2012.